Aegis

Blockchain Smart Contract Vulnerability Scanner

A lightweight security analysis tool designed to **detect common vulnerabilities in Ethereum smart contracts** before deployment.

The Problem

Smart contracts manage **millions of dollars** in blockchain ecosystems — but small coding mistakes can lead to massive exploits.

Reentrancy Attacks

Malicious contracts drain funds by repeatedly calling back into vulnerable functions.

Integer Overflow

Arithmetic errors cause unexpected behavior and exploitable logic flaws.

Unauthorized Access

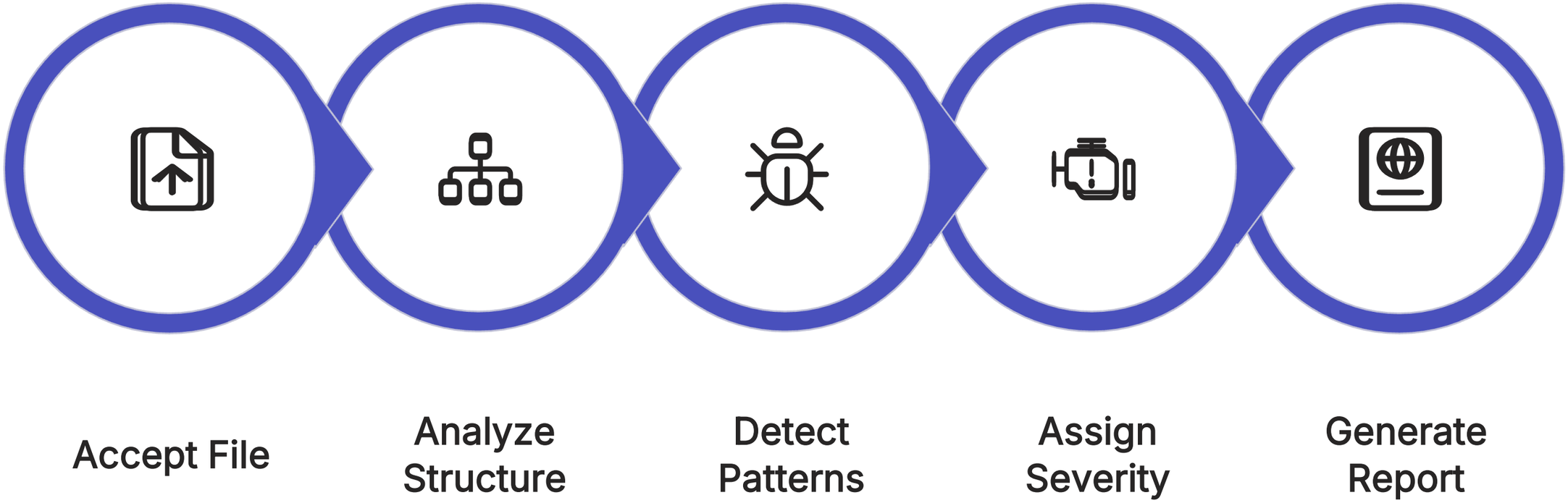
Missing access controls expose critical contract functions.

Weak Randomness

Predictable on-chain randomness enables manipulation of outcomes.

Project Solution

Aegis is a **static analysis security scanner** that automates vulnerability detection before contracts go live.



The pipeline transforms raw Solidity code into an actionable security report — catching risks before they become exploits.

System Architecture

Aegis is built on **four core modules**, each handling a distinct phase of the analysis pipeline.

1

Smart Contract Input

Upload `.sol` files, validate format, and prepare for analysis.

2

Vulnerability Detection Engine

Core analysis using static code analysis, pattern matching, regex, and optional AST parsing.

3

Blockchain Simulation

Simulate transactions and demonstrate potential attack scenarios under malicious conditions.

4

Security Report Generator

Produces structured reports with vulnerability type, severity, code location, and fixes.

Vulnerability Detection Engine

Detectable Vulnerabilities

- Reentrancy attacks
- Integer overflow / underflow
- `tx.origin` authentication misuse
- Unprotected `selfdestruct`

Analysis Methods

- Static Code Analysis
- Pattern Matching
- Regex Detection
- AST Parsing

Security Report Output

Every scan produces a **structured vulnerability report** with everything a developer needs to act.

Vulnerability	Severity	Location	Recommended Fix
Reentrancy	High	Line 45	Use ReentrancyGuard
Integer Overflow	Medium	Line 78	Use SafeMath library
tx.origin Misuse	High	Line 22	Replace with msg.sender
Unprotected selfdestruct	Critical	Line 91	Add access modifier

Tech Stack

Python

Core scanning engine powering all analysis logic.

Solidity

Target smart contract language for analysis.

Flask

Optional web interface for browser-based scanning.

Regex / AST Parsing

Precise vulnerability pattern detection in source code.

Ganache

Local blockchain simulation for attack scenario testing.

Why Aegis Matters

Real-World Impact

Prevents financial losses caused by undetected smart contract vulnerabilities before deployment.

Key Concepts Demonstrated

Static analysis, vulnerability detection algorithms, blockchain simulation, and secure software development.

Developer Empowerment

Gives developers actionable, specific guidance to fix issues — not just detect them.

Future Improvements

Aegis is built with extensibility in mind — a strong foundation for continued growth.

Slither / Mythril Integration

Leverage industry-standard engines for deeper analysis.

More Vulnerability Types

Expand detection coverage beyond current patterns.

Interactive Web Dashboard

Visual interface for uploading contracts and reviewing reports.

CI/CD Integration

Automate security scanning in smart contract deployment pipelines.

Project Aegis

Secure Contracts. Before They Ship.

A student cybersecurity project focused on **blockchain smart contract security analysis** — demonstrating that proactive scanning can prevent the next major exploit.

Built to detect reentrancy, overflow, unauthorized access, and weak randomness vulnerabilities in Solidity smart contracts using static analysis.